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B. E. (Third Semester) Examination, Nov.-Dec. 2016

(New Scheme)

(EEE Branch)

MATHEMATICS-III

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) from each question is compulsory and carries 2 marks. Attempt any two parts from (b), (c) & (d) of each question and carries 7 marks each.*

Unit-I

1. (a) A function $f(x)$ defined in $(-\pi, \pi)$ can be expanded into Fourier series containing both sine and cosine terms. 2

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(b) Prove that :

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty}(-1)^n \frac{\cos nx}{n^2} - \pi < x < \pi$$

Hence show that :

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(i) $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$

(ii) $\sum \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}$

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(c) Expand the function $f(x) = x \sin x$ as a Fourier series in the interval $-\pi < x < \pi$.

Deduce that :

$$\frac{1}{1\cdot 3} - \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} - \frac{1}{7\cdot 9} \ldots\ldots = \frac{1}{4}(\pi - 2)$$

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(d) Obtain the constant term and the coefficients of the first sine and cosine terms in the Fourier expansion of y as given in the following table :

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x :	0	1	2	3	4	5
y :	9	18	24	28	26	20

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Unit-II

(a) If $f(t)$ is a periodic function with period T write down an expression for the Laplace transform of $f(t)$. 2

(b) (i) Find the Laplace transform of the function : 7

$$\frac{1-\cos t}{t}$$

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(ii) $f(t)=\begin{cases}\cos(t-2\pi/3) & , \quad t>2\pi/3 \\ 0 & , \quad t<2\pi/3\end{cases}$

(c) Apply convolution theorem to evaluate : 7

$$L^{-1}\frac{s^2}{(s^2+a^2)(s^2+b^2)}$$

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(d) Solve the differential equation by applying Laplace transform method : 7

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$$ty''+2y'+ty=\cos t$$

given that

$$y(0)=1$$

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Unit-III

3. (a) Form a partial differential equation : 2

$$z=f(x+at)+g(x-at)$$

(b) Solve the partial differential equation : 7

$$x^2(y-z)p+y^2(z-x)q=z^2(x-y)$$

(c) Solve : 7

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$$\left[D^2-DD'-2D'^2\right]z=(y-1)e^x$$

(d) Find a solution of the equation : 7

$$\frac{\partial^2 u}{\partial x^2}=\frac{\partial u}{\partial y}+2u,$$ in the form

$u=f(x)\cdot g(y)$. Solve that equation subject to the

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conditions $u=0$ and $\frac{\partial u}{\partial x}=\left(1+e^{-3y}\right)$, when $x=0$

for all values of y.

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Unit-IV

4. (a) State Cauchy's residue theorem. 2

(b) If $f(z)$ is an analytic function with constant modulus, show that $f(z)$ is constant. 7

(c) Evaluate : 7

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$$\int_C \frac{\sin \pi z^2+\cos \pi z^2}{(z-1)^2(z-2)}dz$$

where C is the circle $|z|=3$.

(d) Evaluate : 7

$$\int_{-\infty}^{\infty} \frac{x^2 dx}{(x^2+2)(x^2+4)}$$

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Unit-V

5. (a) Evaluate the z-transform of sequance : 2

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$$f(k)=\sum_{k=0}^{\infty} 2^k \sum_{k=0}^{\infty} 3^k$$

(b) (i) Find the z-transform of :

$$c^k \cos \alpha k,\ k \geq 0$$

(ii) Find the z-transform of : 7

$$c^k \cosh \alpha k,\ k \geq 0$$

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(c) (i) Use convolution theorem to evaluate : 7

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$$z^{-1}\left\{\frac{z^2}{(z-a)(z-b)}\right\}$$

(ii) Use Residue method to evaluate

$$z^{-1}\left\{\frac{9z^3}{(3z-1)^2(z-2)}\right\}$$

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(d) Solve the difference equation :

$$y_{k+3} - 3y_{k+2} + 3y_{k+1} - y_k = u(k)$$

$$y(0) = y(1) = y(2) = 0$$

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by z transfer. 7

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